LAPORAN PRAKTIKUM
ALGORITMA DAN PEMROGRAMAN 1
MODUL 3
“Fungsi”

DISUSUN OLEH:

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PROGRAM STUDI S1 TEKNIK INFORMATIKA

FAKULTAS INFORMATIKA

TELKOM UNIVERSITY PURWOKERTO

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DASAR TEORI

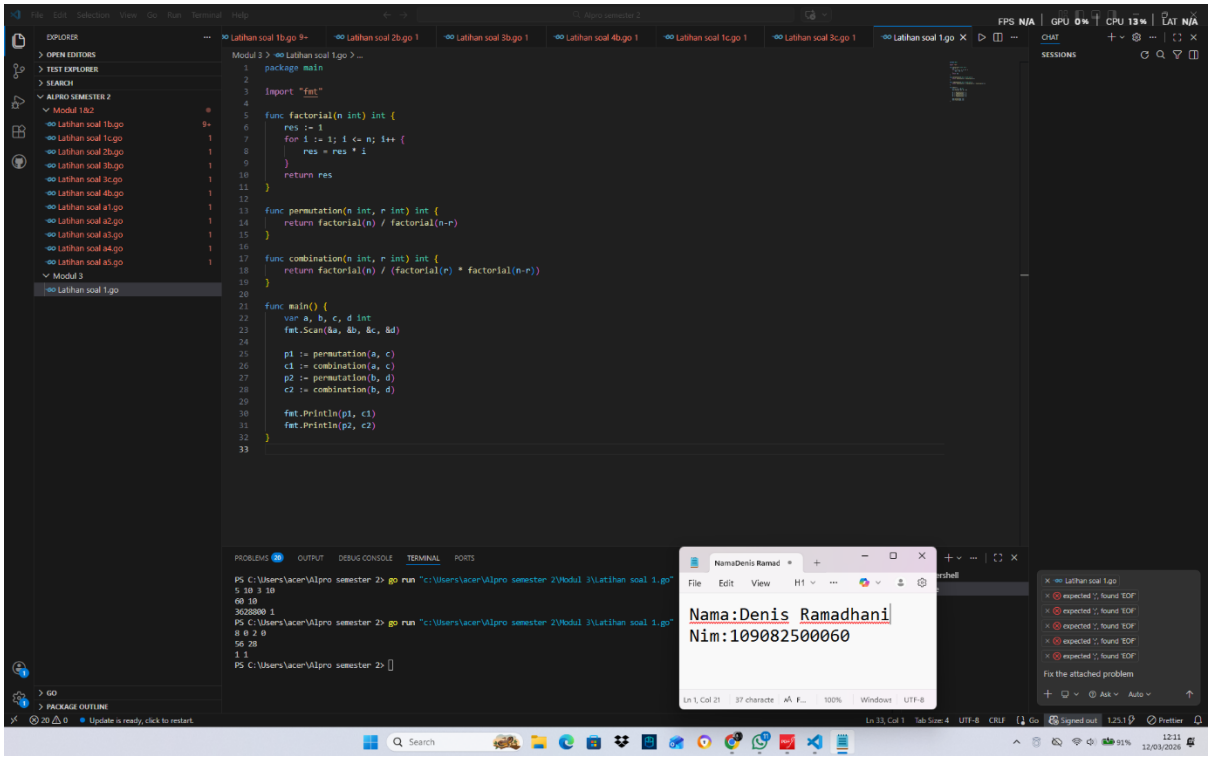
SOAL

1.

Source Code:

```
package main

import "fmt"

func factorial(n int) int {
   res := 1
   for i := 1; i <= n; i++ {
      res = res * i
   }
   return res
}

func permutation(n int, r int) int {
   return factorial(n) / factorial(n-r)
}

func combination(n int, r int) int {
   return factorial(n) / (factorial(r) * factorial(n-r))
}

func main() {
   var a, b, c, d int
   fmt.Scan(&a, &b, &c, &d)

   p1 := permutation(a, c)
   c1 := combination(a, c)
   p2 := permutation(b, d)
   c2 := combination(b, d)

   fmt.Println(p1, c1)
   fmt.Println(p2, c2)
}
```

Output:



Deskripsi Program:

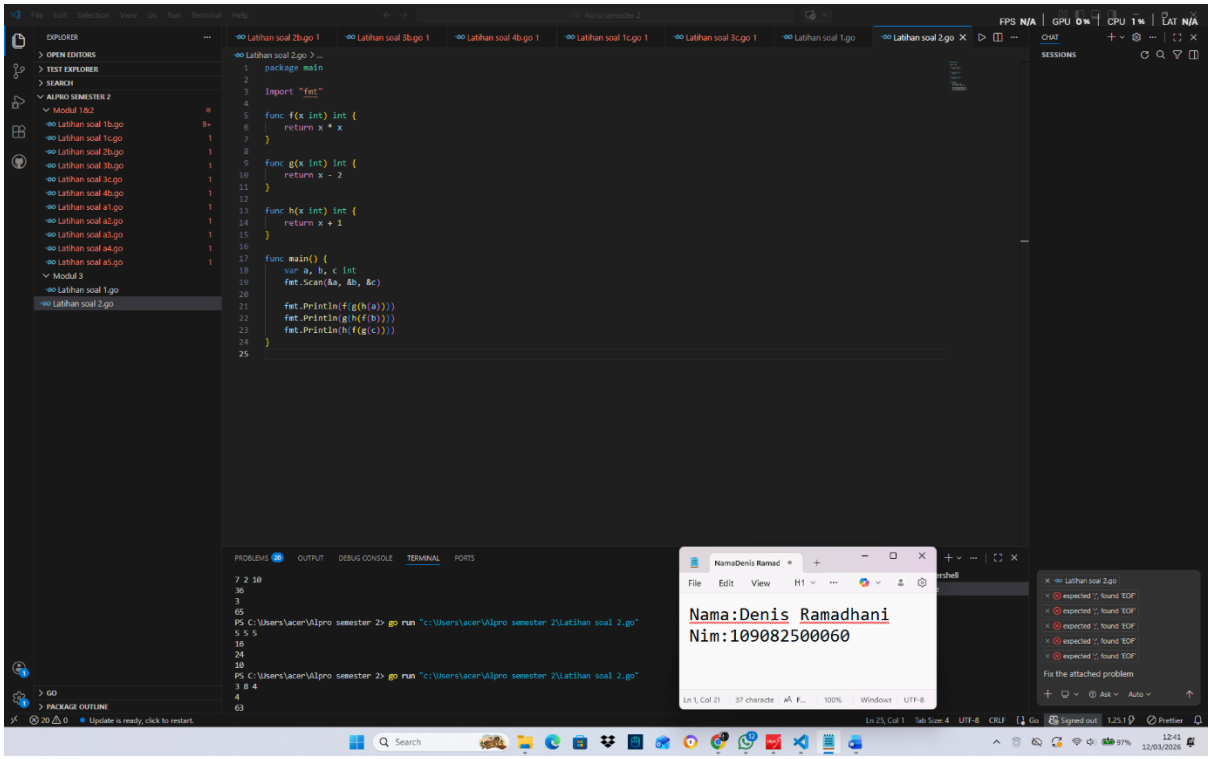
Menghitung nilai permutasi dan kombinasi dari input angka menggunakan fungsi faktorial.

2

Source Code:

```
package main

import "fmt"

func f(x int) int {
   return x * x
}

func g(x int) int {
   return x - 2
}

func h(x int) int {
   return x + 1
}

func main() {
   var a, b, c int
   fmt.Scan(&a, &b, &c)

   fmt.Println(f(g(h(a))))
   fmt.Println(g(h(f(b))))
   fmt.Println(h(f(g(c))))
}
```

Output:



Deskripsi Program:

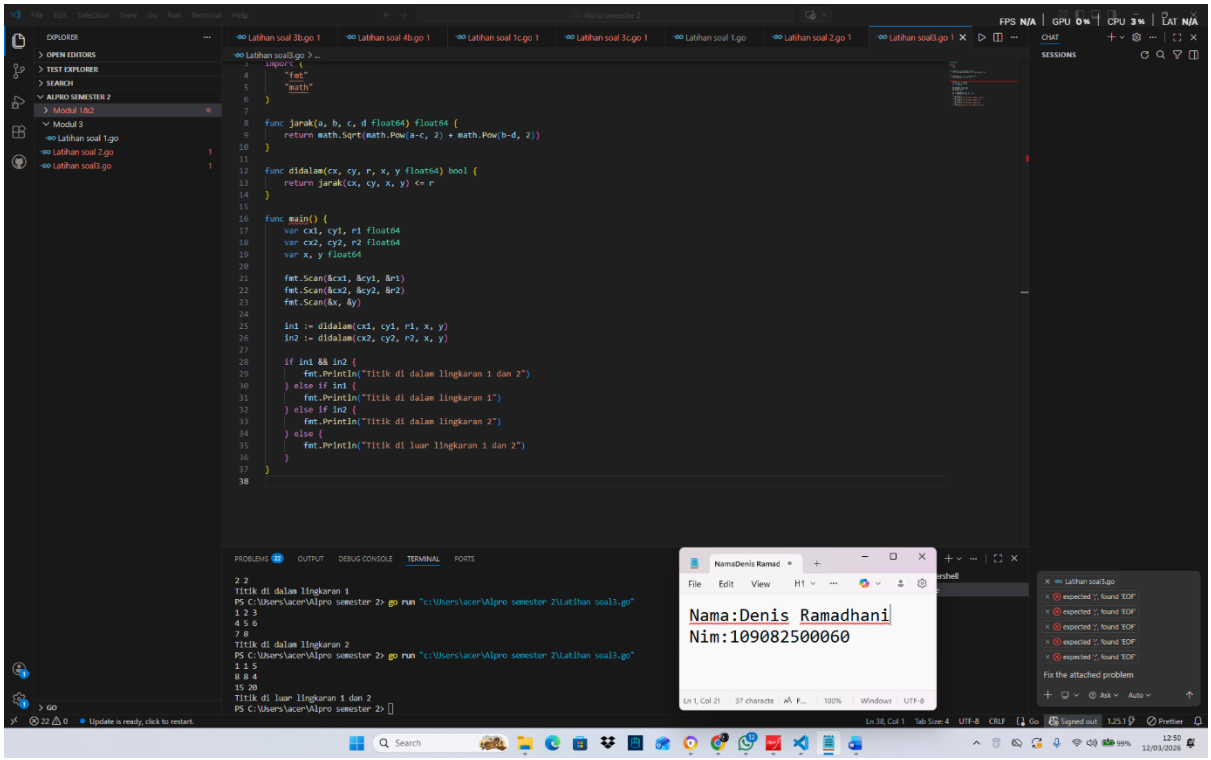
Menghitung hasil komposisi matematika dari tiga fungsi berbeda ($f, g,$ dan h) secara berurutan.

3

Source Code:

```
package main

import (
   "fmt"
   "math"
)

func jarak(a, b, c, d float64) float64 {
   return math.Sqrt(math.Pow(a-c, 2) + math.Pow(b-d, 2))
}

func didalam(cx, cy, r, x, y float64) bool {
   return jarak(cx, cy, x, y) <= r
}

func main() {
   var cx1, cy1, r1 float64
   var cx2, cy2, r2 float64
   var x, y float64

   fmt.Scan(&cx1, &cy1, &r1)
   fmt.Scan(&cx2, &cy2, &r2)
   fmt.Scan(&x, &y)

   in1 := didalam(cx1, cy1, r1, x, y)
   in2 := didalam(cx2, cy2, r2, x, y)

   if in1 && in2 {
      fmt.Println("Titik di dalam lingkaran 1 dan 2")
   } else if in1 {
      fmt.Println("Titik di dalam lingkaran 1")
   } else if in2 {
      fmt.Println("Titik di dalam lingkaran 2")
   } else {
      fmt.Println("Titik di luar lingkaran 1 dan 2")
   }
}
```

Output:



Deskripsi Program:

Mengecek posisi koordinat titik terhadap dua lingkaran menggunakan rumus jarak Euclidean dan fungsi Boolean.

DAFTAR PUSTAKA